

Toward Size-Robust Zero-Shot Search: Expert Imitation, Iteration Scaling, and What the Ablations Attribute (v7)

neuroMCTS project · Research meeting · July 2026

Five literature lessons shaped v7: imitate, scale iterations, normalize features, align with propagation, randomize sizes

Teacher: propagation-impact expert (fail-first probing) replaces self-play — learn2branch pattern

Architecture: weight-shared recurrent GNN → message-passing iterations scalable at test time (NeuroSAT)

Data: sizes $m \sim U\{6..20\}$, all instances fresh; strict zero-shot: 40×100 never seen in training

v7 perfects 10×25 — the first non-exact 100/100 on the hard family — but regresses at 20×50

10×25: feas 100% / sol 100% at 1.52 s (v6: 99.3/100 — all 7 budget FPs now proven infeasible)

20×50: sol 5.5% vs v6 9.6% — regression, attributed by ablation to the imitation phase

40×100 zero-shot: 2.25%, identical to v6 (trained on that size) — extrapolation is pump-bound

Matched ablations: iteration count is innocent; the imitation phase caused the regression

Imitation ckpt, 16 iters: 5.50% | imitation ckpt, 8 iters (trained regime): 5.50%

Pretrain-only ckpt, same budgets: 8.50% — the teacher, not the iterations, is the cause

Third independent failure of open-loop policy training (v6 self-play, v6 ordering-REINFORCE, v7 imitation)

The fail-first asymmetry: one ordering policy cannot serve both proofs and solutions

Fail-first teacher $\rightarrow$ proof trees shrink $\rightarrow$ 10 $\times$ 25 infeasibility proofs 48 $\rightarrow$ 55, FP 7 $\rightarrow$ 0

Same teacher $\rightarrow$ solution search degrades at 20 $\times$ 50 (succeed-first guidance needed there)

Classical CP wisdom, now measured in our pipeline — ordering policy must be role-specific

Test-time iteration scaling is a null result: extrapolation stays pump-bound

40×100 zero-shot: 8 vs 32 message-passing rounds → identical outcomes (80.2% / 2.25%)

All 9 solutions from the kernel pump in both configurations

No inference-regime adjustment (budget, iterations) bridges the hardness boundary

Conclusion: learned ordering should target proofs; solutions need lattice enumeration

Deploy per size: imitation checkpoint where proofs dominate (small), pretrained where search dominates

Next: no-good DFS (conflict learning), then LLL-reduced kernel-basis enumeration with learned guidance

If policy training returns: optimize proof-tree size on infeasible instances — the one transferable gain